

1 Consider a demand-paging virtual memory system with a memory of four frames. A process has seven pages: 1,2,3,4,5,6,7, it runs with a memory reference string:

1, 2, 3, 6, 4, 7, 3, 2, 1, 4, 7, 5, 6, 5, 2, 1

a) Draw the frames that illustrate the execution of these pages using FIFO, LRU algorithms.

b) Show the number of page faults for each algorithms.

c) Which four pages that remain in the frames at last for each algorithms.

2 A file is made up of 128-byte fix-sized logical records and stored on the disk in the unit of the block that is of 1024 bytes. The size of the file is 10240 bytes. Physical I/O operations transfer data on the disk into an OS buffer in main memory, in terms of 1024-byte block. If a process issues *read* requests to read the file's records in the sequential access manner, what is the percentage of the *read* requests that will result in I/O operations?

3

a) Consider the following page table:

		frame number	
		0	
		1	
Page 0	0	2	Page 2
Page 1	1	3	Page 0
Page 2	2	4	
		5	Page 3
		6	Page 1
		7	

logical memory

page table

physical memory

The page size is 1024 bytes. What are the physical addresses for the following logical addresses?

(1) 230 (2) 1129 (3) 2104 (4) 4094

b) Consider the following segment table:

Segment	Base	Length
0	215	500
1	2404	200
2	90	45
3	1347	250

What are the physical addresses for the following logical addresses?

(1) <0, 480> (2) <1, 143> (3) <2, 16> (4) <3, 400>